Week 4: Medium Access Control (MAC) Layer Principles

4.1 The Role of the MAC Layer in Wireless Networks

4.1.1 The Architectural Abstraction

In network architecture, the Data Link Layer (Layer 2) is split into two distinct functional sublayers:

1. **Logical Link Control (LLC):** Manages flow control, upper-layer multiplexing, and error checking.
2. **Medium Access Control (MAC):** Sits directly above the Physical Layer (PHY) to regulate exactly how devices share and access the physical network medium.

4.1.2 The Shared Broadcast Domain: Wired vs. Wireless

How wired and wireless mediums handle data traffic:

+ **Wired Domain (Switched Ethernet):** Modern wired systems connect computers to full-duplex switches using dedicated, isolated physical lines. Because traffic paths are physically separated, collision domains are completely isolated, making media contention a non-issue.
+ **Wireless Domain (Radio Frequency Space):** Wireless networks operate inside an unguided, open radio broadcast domain. The airwaves are entirely shared. If two devices within radio range transmit electromagnetic signals over the same frequency channel at the same time, the signals mix destructively at the receiver. This event is a data packet collision, resulting in corrupted, unreadable noise.

4.1.3 Core Functional Responsibilities of the Wireless MAC

Because the physical radio medium introduces intense interference, signal fading, and environmental barriers, the wireless MAC layer must guarantee reliable data delivery through three primary functions:

* **Channel Access Control:** Arbitrating which node transmits and exactly when they transmit to maximize channel efficiency and maintain node fairness.
* **Framing and Physical Addressing:** Packaging network-layer packets into structured MAC frames, embedding the unique physical source and destination Hardware (MAC) addresses.
* **Link-Layer Error Mitigation:** Wireless link error rates are orders of magnitude higher than wired lines due to ambient noise. The MAC layer handles rapid, localized error detection and frame retransmissions (via Link-Layer Acknowledgments or ACKs) without waiting for slower, higher-layer protocols like TCP to time out.

4.2 Managing Collisions — The CSMA/CA Protocol

4.2.1 Why Wireless Cannot Detect Collisions (The Death of CSMA/CD)

Traditional wired Ethernet uses CSMA/CD (Carrier Sense Multiple Access with Collision Detection). When a wired node transmits, it monitors the physical voltage on the cable. If the voltage spikes, the node immediately detects a collision, ceases transmission, and backs off.

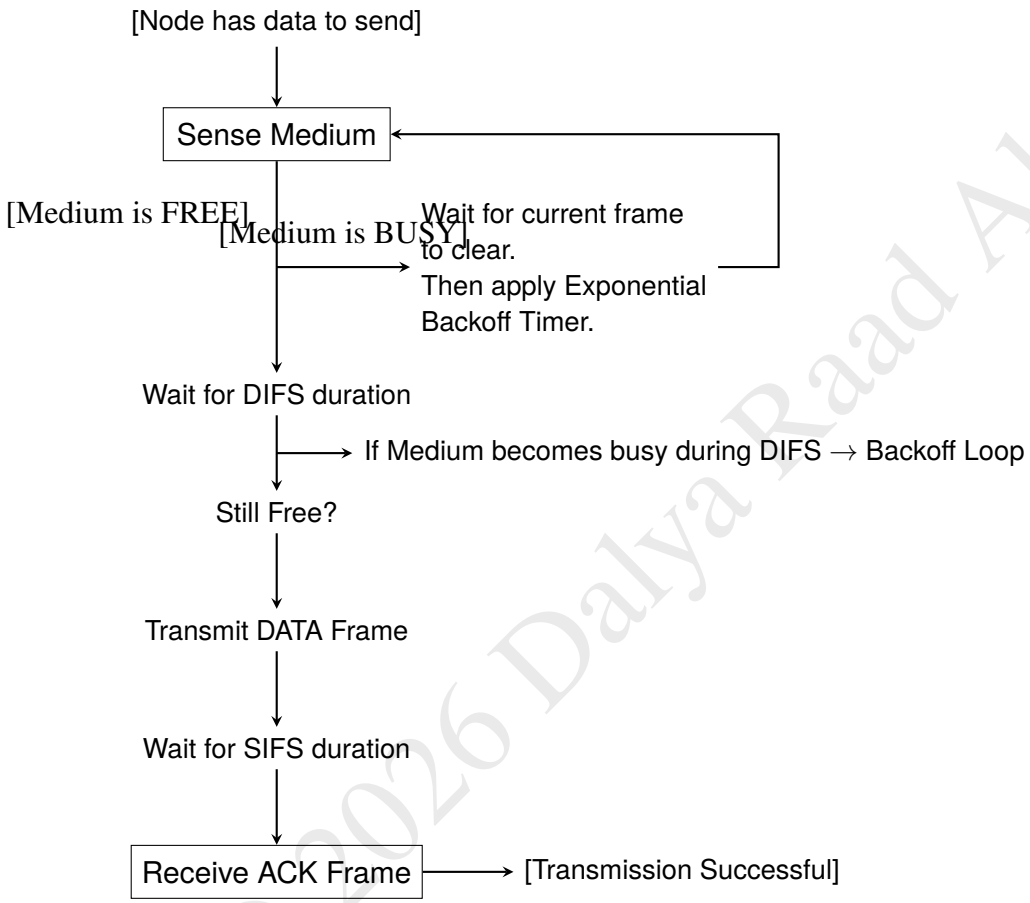
Wireless networks cannot do this. A wireless node's local transmission power completely drowns out any incoming radio signal at its own antenna. A wireless transceiver is functionally "deaf" to the outside world while it is actively speaking. Because it cannot detect a collision mid-transmission, a wireless node would blindly transmit a corrupted frame until the very end, wasting massive amounts

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of airtime and battery energy. Therefore, wireless networks abandon Collision Detection in favor of Collision Avoidance (CSMA/CA).

4.2.2 The Detailed Protocol Mechanics of CSMA/CA

Under Carrier Sense Multiple Access with Collision Avoidance (used in IEEE 802.11 Wi-Fi networks), a station avoids collisions by applying a strict wait-and-sense routine before opening its transmitter.



* **Exponential Backoff Timer Mechanics:** When a node calculates a backoff, it samples: Backoff Time $=$ Random$(0, CW) \times$ Slot Time. If a transaction fails to generate an ACK, the Contention Window range doubles ($CW_{min} \rightarrow 2 \times CW_{min}$) up to CW_{max} to stagger congested packets.
* **Interframe Space Priority Hierarchy:** The configuration enforces strict structural priorities: **SIFS (Short IFS)** for control frames (ACK, CTS) and **DIFS** for standard data frames. Because SIFS $<$ DIFS, the ACK frame grabs the link first, blocking outer waiting data streams.

4.3 Structural Wireless Anomalies

4.3.1 The Hidden Terminal Problem

+ **Scenario:** Two transmitting stations (A and C) are placed outside each other's functional radio coverage boundaries, but interact with a common receiver (B) situated within mutual reach.
+ **Failure Mechanic:** A checks the channel, senses silence, and transmits to B. Simultaneously, C senses the channel. Because C is out of radio range of A, C hears absolutely nothing on the airwaves, falsely assumes the channel is free, and transmits to B.
+ **Consequence:** Both signals converge simultaneously at Receiver B, causing a destructive collision. The streams drop into unreadable noise, causing packet drop errors.

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4.3.2 The Exposed Terminal Problem

+ **Scenario:** Node B is transmitting to Node A. Node C sits close enough to hear B, but wants to send a separate frame to an independent target D, located completely out of reach of B.
+ **Failure Mechanic:** C executes physical carrier sensing, hears B speaking, and aggressively defers its transmission to avoid causing a collision.
+ **Consequence:** This delay is entirely unnecessary. C's transmission to D would not affect A's reception of B's signal. This topological flaw limits spatial reuse capacity.

4.3.3 The Protocol Resolution: Virtual Carrier Sensing (RTS/CTS & NAV)

To resolve both anomalies simultaneously, the MAC layer provides an optional four-way control handshake routine:

1. **RTS (Request to Send):** The sender transmits a short packet embedded with a precise microsecond duration field representing the window required to pass the data payload and final verification ACK.
2. **CTS (Clear to Send):** The destination responds with a matching CTS frame echoing this duration metric.
3. **NAV (Network Allocation Vector):** Third-party stations overhearing the RTS or CTS extract the duration parameter and configure an internal virtual clock countdown. The node enforces silence until the NAV hits zero, shielding the transmission stream even if physical radio sensors detect silence.

4.4 Near-Far Interference & Power Control

* **The Phenomenon:** Electromagnetic paths attenuate quickly over space. A station positioned extremely near a central receiver will project a massive received signal profile that can drown out the deeply attenuated, faint signal profile arriving from a distant edge node.
* **System Impact:** This is highly critical in CDMA paths where all users share identical bands. The immense near-field signature bleeds past orthogonal separation metrics, completely blinding the channel base station to outer ring devices.
* **The MAC Solution:** Real-time, closed-loop power control. The base station calculates incoming signal metrics (SINR) and streams fast-response power scaling flags over the MAC channel to step up edge node power thresholds while cutting the power levels of near-field devices.

4.5 Analysis of Contention-Based vs. Contention-Free Access

+ **Contention-Based Access:** Nodes compete stochastically for transmission slots on a first-come, first-served architecture (e.g., CSMA/CA, ALOHA). Highly efficient and lean for irregular, low-density bursty traffic demands, but breaks down under heavy traffic saturation due to excessive backoff loops.
+ **Contention-Free Access:** Central network orchestrators assign deterministic, non-overlapping coordinate blocks to each active terminal using TDMA, FDMA, or Polling maps. Eliminates multi-user collisions entirely, providing reliable QoS guarantees for live voice or industrial loops, but carries strict synchronization clock burdens.

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